

Xi Emmett Peng

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Education

Research Center for Eco-Enviromental Sciences, CAS <i>Ph.D. in Ecology, advised by Prof. Ye Deng</i>	2019/09 – 2026/01 <i>Beijing, China</i>
KU Leuven <i>Visiting Ph.D. student, co-supervised by Dr. Karoline Faust</i>	2024/09 – 2025/09 <i>Leuven, Belgium</i>
China Agricultural University <i>Bachelor of Science, Biological Sciences</i>	2015/09 – 2019/06 <i>Beijing, China</i>

Research Experience

Q-SCOPE development <i>RCEES, CAS & KU Leuven</i>	2024.01 – Present <i>Beijing, China & Leuven Belgium</i>
- Developed Q-SCOPE (Quorum-Sensing Communication Link Predictor) as a containerized web application to retain profile-level evidence, apply versioned component criteria to assign quorum-sensing-sending and -receiving potential, and predict pathway-specific potential communication links. - Demo version: https://www.emmettnotebook.com/Q-SCOPE - Publication: <i>Q-SCOPE profiles quorum-sensing components and capabilities to predict potential communication links in microbial communities</i> (In submission)	
Metabolic complementarity network modeling <i>Research Center for Eco-Environmental Sciences, Chinese Academy of Sciences</i>	2021.09 – 2024.06 <i>Beijing, China</i>
- Developed a metagenomics workflow combining co-occurrence and metabolic complementarity networks to identify microbial interactions in thermophilic communities. - Revealed that high temperatures strengthen metabolic cooperation, particularly between distantly related microbes and species with large differences in genome size. - Publications: <i>Metabolic interdependencies in thermophilic communities are revealed using co-occurrence and complementarity networks</i> (Nature Communications) & <i>iNAP 2.0: Harnessing metabolic complementarity in microbial network analysis</i> (iMeta)	
Scientific training <i>Univeristy of Chinese Academy of Sciences & RCEES, CAS</i>	2019.09 – 2021.09 <i>Beijing, China</i>
- Took courses for basic research skills for microbial ecology, molecular biology, bioinformatics and biostatistics. - Participated in research projects related to 16S rRNA gene sequencing data analysis, molecular ecological network analysis and metagenome sequencing data analysis.	

Selected Publications

Xi Peng, Shang Wang, Miaoxiao Wang, Kai Feng, Qing He, Xingsheng Yang, Weiguo Hou, Fangru Li, Yuxiang Zhao, Baolan Hu, Xiao Zou, Ye Deng*. (2024) Metabolic interdependencies in thermophilic communities are revealed using co-occurrence and complementarity networks. *Nature Communications*. Doi: <https://doi.org/10.1038/s41467-024-52532-x>

Xi Peng, Kai Feng*, Xingsheng Yang, Qing He, Bo Zhao, Tong Li, Shang Wang, Ye Deng*. (2024) iNAP 2.0: Harnessing metabolic complementarity in microbial network analysis. *iMeta*. Doi: <https://doi.org/10.1002/imt2.235>

Xi Peng, Xingsheng Yang, Danrui Wang, Bo Zhao, Kai Feng, Qing He, Shang Wang, Ye Deng*.
(2026) From short to long: The impact of read length on metagenome assembly and binning.
Methods in Ecology and Evolution. Doi: <https://doi.org/10.1111/2041-210x.70306>

For a complete list of publications, see: <https://www.emmettnotebook.com/publications/>.

Awards & Honors

the Grand Award of the President Scholarship of the Chinese Academy of Sciences	
<i>Chinese Academy of Sciences</i>	2025
China National Scholarship for Doctoral Students	
<i>Ministry of Education of the People's Republic of China</i>	2025
CSC Scholarship	
<i>China Scholarship Council</i>	2024
Outstanding contribution in Scientific Innovation	
<i>Wiley China Excellent Author Program</i>	2024
The special prize scholarship	
<i>Research Center for Eco-Environmental Sciences, Chinese Academy of Sciences</i>	2024
The first prize scholarship (4 consecutive prizes)	
<i>Research Center for Eco-Environmental Sciences, Chinese Academy of Sciences</i>	2020, 2021, 2022, 2023

Skills and Expertise

Molecular biology skills: Peng is proficient in standard microorganism-laboratory techniques, including DNA extraction, PCR, gel electrophoresis, aseptic technique, and microbial culturing

Bioinformatic skills: Peng is well-trained in comprehensive metagenomic analysis, with expertise spanning amplicon and shotgun sequencing. His contributions include developing and maintaining the Galaxy Platform for amplicon/functional gene analysis (DMAP) and the Integrated Network Analysis Pipeline (iNAP), showcasing his skills in coding and bioinformatics infrastructure development. Proficient in the standard metagenomic analysis workflow, Peng adeptly utilizes quality control (Trimmomatic, NextPolish), read assembly (MEGAHIT, metaSPAdes, IDBA-UD, OPERA-MS), and genome binning software (metaWRAP, metaBAT2, MAXBIN2, concoct, VAMB), among others.

Computer skills: A self-proclaimed computer enthusiast and emerging geek, Peng boasts advanced skills in Linux (Shell/Bash), Python, and R programming. He is keenly interested in AI technologies and prompt engineering, with practical experience in various AI tools such as ChatGPT and MidJourney. His expertise extends to statistical visualization, where he excels in R, Cytoscape, and Gephi, illustrating his versatile computational abilities.

Languages: Chinese Mandarin (native), English (fluent).